How does demand from U.S. power plants shape investment opportunities and risks in the coal industry?

By Bennett Stooksbury, Jayden Bayuga, and Jeena Huynh

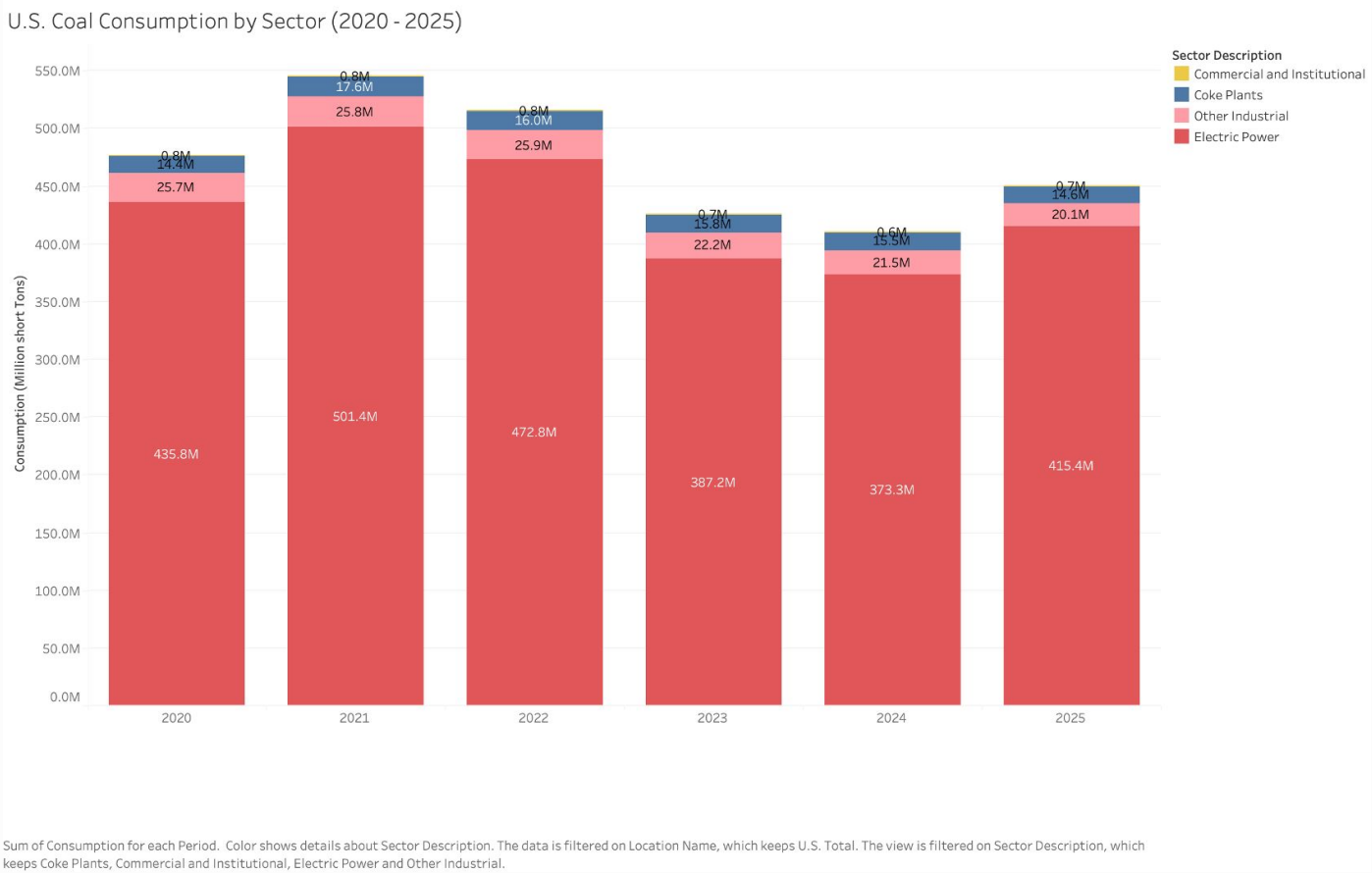
Findings:

We recommend a selective overweight in coal suppliers serving U.S. power plants over the next 6–12 months. The rebound in coal consumption and declining stockpiles could lead power plants to increase purchases, supporting additional orders for suppliers. This opportunity depends on demand remaining strong and customers rebuilding their inventories.

Main Takeaways:

Electric Power accounts for:

92.2% U.S. Coal Use



Sum of Consumption for each Period. Color shows details about Sector Description. The data is filtered on Location Name, which keeps U.S. Total. The view is filtered on Sector Description, which keeps Coke Plants, Commercial and Institutional, Electric Power and Other Industrial.

Main Takeaways:

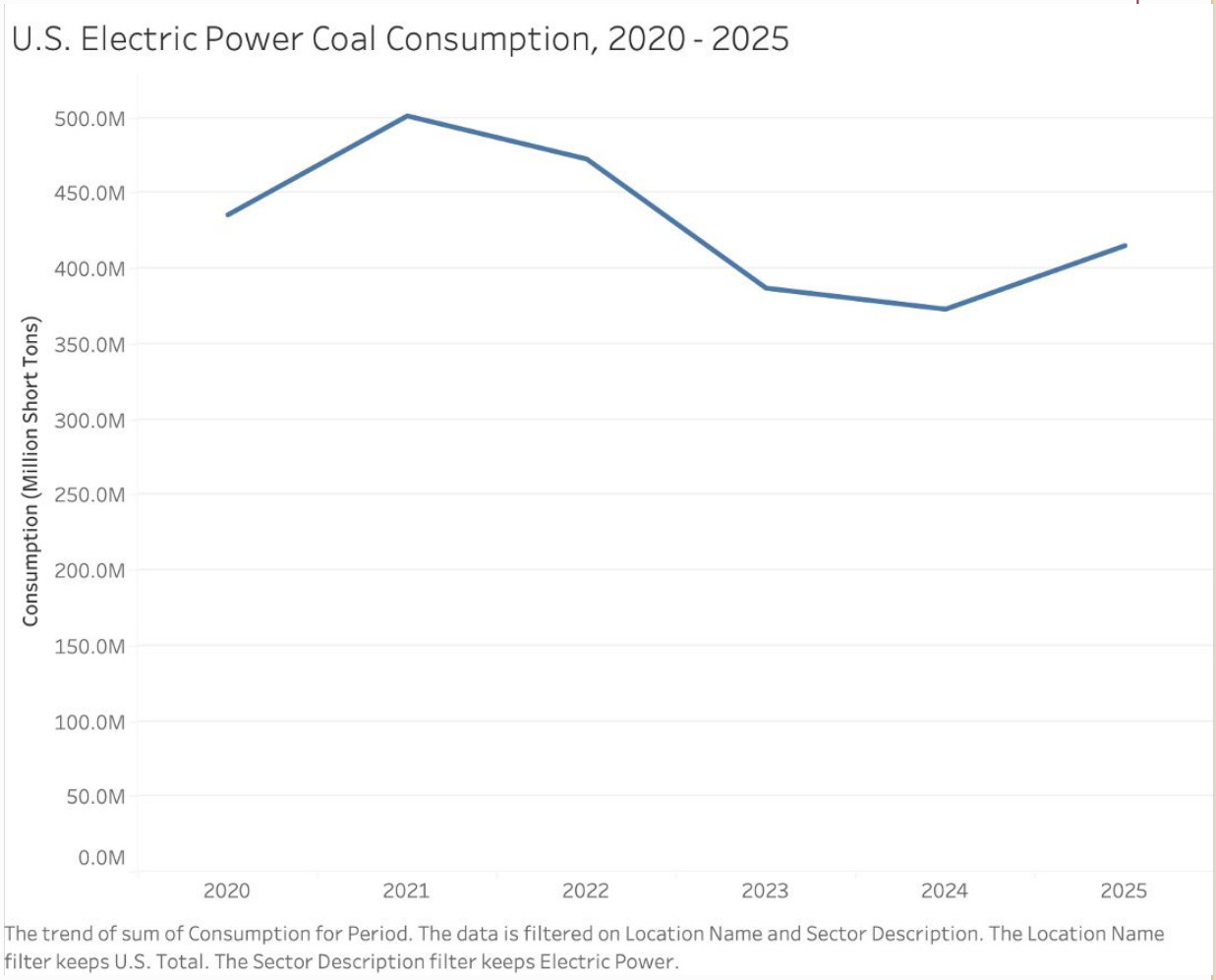
11.3% Increase

Consumption in 2025 from 2024

17.2% Decrease

Consumption in 2025 from 2024

Potential Investment Opportunity: Continued growth in electric power coal consumption could increase demand for coal suppliers, supporting future sales and revenue.



Electric Power Coal Deliveries vs. Consumption, 2020–2025



The trends of Consumption and Receipts for Period. Color shows details about Consumption and Receipts. The data is filtered on Location Name and Sector Description. The Location Name filter keeps U.S. Total. The Sector Description filter keeps Electric Power.

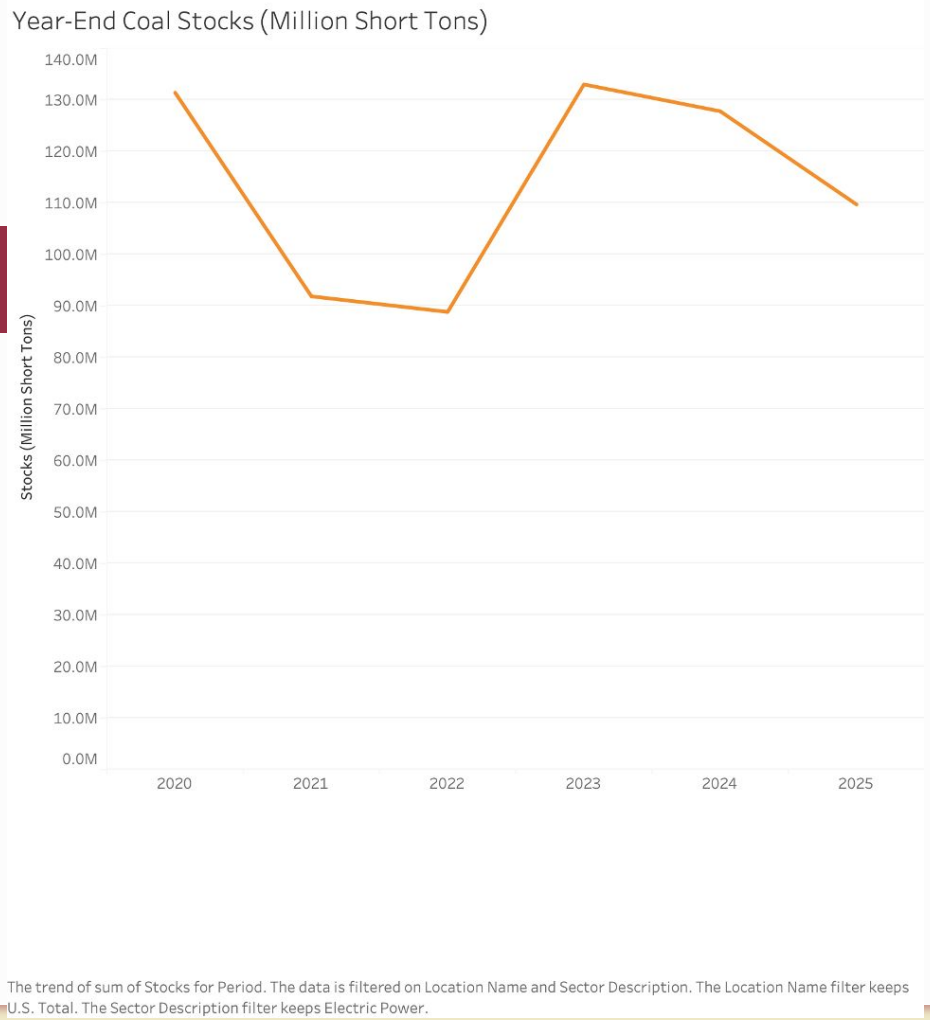
Main Takeaways:

Consumption Greater than **Deliveries** in **2025**

2025: 415.M tons consumed vs. 389.9M tons received, **25.5M-ton gap.**

Declining stockpiles suggest power plants **reliance** on stored coal to meet **demand**.

If **demand Stays Strong**, Power Plans may **increase coal purchase** to replenish stockpiles, **benefiting** coal producers.



Main Takeaways:

Electric Power had a:

14.2% Decline

Stockpiles in 2025

Declining stockpiles, combined with consumption exceeding receipts, suggests power plants relied on stored coal to meet demand

Potential Investment Opportunity:
If coal consumption remains strongs, power plants may increase purchases to replenish stockpiles, potentially benefiting coal suppliers.



Main Takeaways:

Average delivered coal prices increased from 2020 to 2023 before declining slightly in 2025.

2020-2023: Prices increased from **$36.13 to $46.96 per short ton.**

2025: Prices decreased to **$45.56 per short ton,** remaining above 2020 levels.

Potential Investment Opportunity: Higher coal prices could support supplier revenue, while rising costs for power plants may reduce coal demand and create investment risks.

Main takeaways:

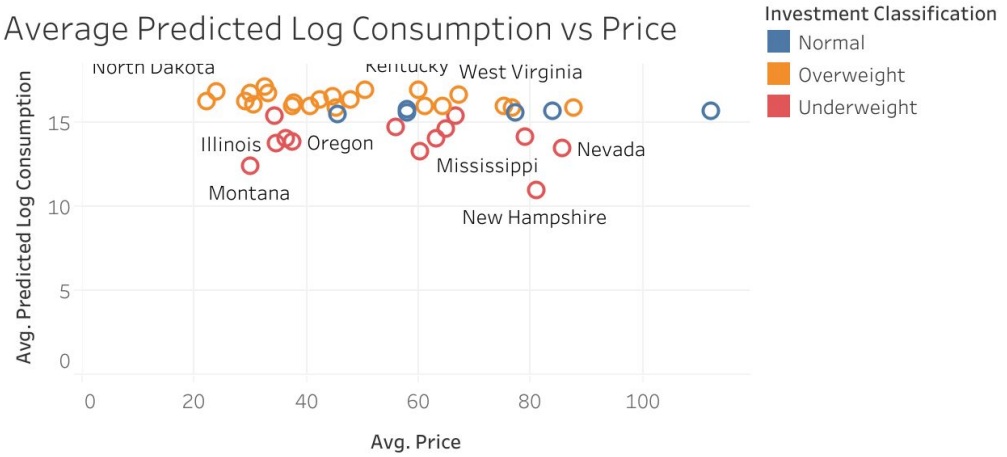
Coal Prices and predicted consumption vary across states, highlighting differences in market conditions.

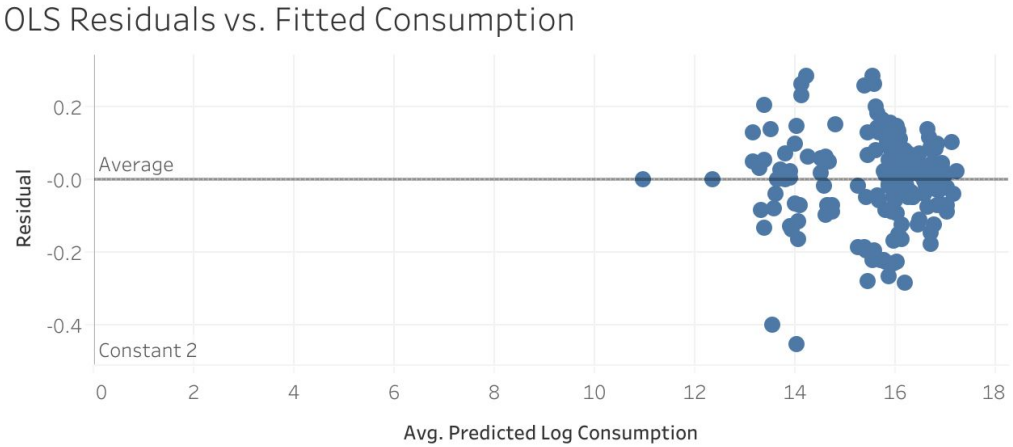
New Hampshire: Approximately **$80 per ton,** with a predicted log consumption of **10.96**, indicating a relatively small coal market.

North Carolina: Higher coal price of approximately **$112 per ton**, demonstrating differences in coal costs across states.

Regression Results: The residual plot can help assess model reliability, although it does not establish prediction accuracy on its own

Investment Implications: Differences in coal demand and prices across states may create opportunities in larger markets, while smaller markets, and higher costs present potential risks.





Possible Risks

Market Risk

1. We found that coal demand is growing in some sectors and recessing in others so we need to be careful about inferring overall demand.

2. We don't have data on the global market of coal which affects the U.S. market

Data/Modeling Risk

1. We had some uncertainties about our data cleaning thus this trickle down to our graphs

2. For our OLS regression we should be wary of autocorrelation which could underestimate uncertainty

Thank you for your time